PostgreSQL Yum Installation: Deploy PostgreSQL on RHEL, CentOS, AlmaLinux & Rocky — PostgreSQL Foundations Series

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PostgreSQL is the backbone of many mission-critical systems — but the way you install it can make or break maintainability, security, and upgrade paths. On Red Hat-based Linux systems (RHEL, CentOS, Rocky, AlmaLinux), **Yum Dnnr** (dnf) is the native package manager manner.

This guide will walk you through the **PostgreSQL Yum-based installation** process and best practices, from repo setup to service startup and basic tuning.

🤔 Why Use Yum for PostgreSQL?

PostgreSQL is included in default RHEL/CentOS/AlmaLinux repos — but those versions are often outdated. The PostgreSQL Global Development Group (PGDG) maintains **official Yum repositories** offering:

- Latest stable PostgreSQL versions
- Consistent packaging across versions
- Extension support (PostGIS, pg_cron, etc.)
- Tools for backups, monitoring, and replication

Using PGDG's Yum repo ensures you stay current without compiling from source.

✅ 1. Prep Your System
Prerequisites:

- RHEL-based distro (RHEL 7/8/9, CentOS 7, Rocky Linux, AlmaLinux)
- Root or sudo privileges
- Internet connectivity (for repo access)

Check your OS version:

```
cat /etc/redhat-release
```

📦 2. Add the Official PostgreSQL Yum Repository

The PostgreSQL Global Development Group provides a convenient .rpm file that auto-configures the repo.

Choose your version (e.g., PostgreSQL 16):

```
sudo dnf install -y https://download.postgresql.org/pub/repos/yum/reporpm
```

Replace EL-9=x86_64 with EL-8, EL-7, etc., based on your system.

Disable built-in PostgreSQL repo:

```
sudo dnf -qy module disable postgresql
```

Why? The built-in repo might only include outdated versions (e.g., PostgreSQL 10 or 12), not the latest.

⬇️ 3. Install PostgreSQL via Yum

Now you can install your desired version.

```
sudo dnf install -y postgresql16-server postgresql16-contrib
```

Available packages (vary by version):

- `postgresql16-server`
- `postgresql16-contrib`
- `postgresql16-devel`
- `postgresql16-libs`
- `postgresql16-docs`

Check installed version:

```
postgres --version
```

🗂 4. Initialize the Database Cluster

Before starting PostgreSQL, you must initialize its data directory.

```
sudo /usr/pgsql-16/bin/postgresql-16-setup initdb
```

This sets up the necessary files in /var/lib/pgsql/16/data.

▶️ 5. Start and Enable the PostgreSQL Service

```
sudo systemctl enable postgresql-16
sudo systemctl start postgresql-16
```

Check status:

```
sudo systemctl status postgresql-16
```

🔐 6. Basic Access and Configuration
Login as postgres:

```
sudo -i -u postgres
psql
```

Create database and user:

```
CREATE DATABASE devdb;
CREATE USER jack WITH PASSWORD 'securepass';
GRANT ALL PRIVILEGES ON DATABASE devdb TO jack;
```

Exit with `\q`.

⚙️ 7. Common Configuration Tweaks
File: /var/lib/pgsql/16/data/postgresql.conf

Change:

```
listen_addresses = '*'
```

File: /var/lib/pgsql/16/data/pg_hba.conf

Add for network access:

```
host    all             all             192.168.1.0/24          md5
```

Reload config:

```
sudo systemctl reload postgresql-16
```

🔄 8. Tips for Managing PostgreSQL with Yum
Upgrade minor versions (safe):

```
sudo dnf update postgresql16*
```

Upgrade major versions:

Use pg_upgrade or pg_dump strategy. The Yum repo installs new major versions **side-by-side**.

View available versions:

```
dnf list available | grep postgresql
```

🧩 9. Extensions & Add-ons

Install extras:

```
sudo dnf install postgresql16-contrib
sudo dnf install postgresql16-pg_stat_statements
sudo dnf install postgresql16-plperl
```

Enable in psql:

```
CREATE EXTENSION pg_stat_statements;
```

📁 10. Yum PostgreSQL Folder Structure

🔁 11. Backup & Restore Examples
Dump entire DB:

```
pg_dumpall -U postgres > alldb.sql
```

Dump single DB:

```
pg_dump -U postgres -f c devdb > devdb.dump
```

Restore:

```
pg_restore -U postgres -d newdb devdb.dump
```

🚨 12. Troubleshooting Tips
PostgreSQL won't start?

- Check logs in /var/lib/pgsql/16/data/pg_log/
- Confirm permissions on data directory
- Run `sudo journalctl -u postgresql-16`

Can't connect?

- Check listen_addresses and pg_hba.conf
- Test with psql -U postgres -h localhost
- Check firewall (firewalld, iptables)

📋 Summary: Yum Installation Flow

```
Add Yum Repo
    |
Install Server & Tools
    |
Init DB Cluster
    |
Start & Enable Service
    |
Configure Access & Network
    |
Use & Extend PostgreSQL
```

💭 Final Thoughts

Installing PostgreSQL via Yum is the **best-practice method** for Red Hat-family Linux systems. It keeps your system:

- Cleanly packaged
- Easy to maintain
- Upgradeable with minimal effort

You can trust the Yum repositories provided by the PostgreSQL community to stay up to date and production-ready.

#postgresql

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